Keeping the Lights On for Everything That Can’t Go Dark

Always-On Infrastructure, the Grid It Depends On, and Who Must Act

Scope: Parts One through Five drafted; Findings and Asks and Part Six headed and held

Lineage: fourth companion in the generation-mix family, following Generation Mix Optimization Under Competing Objectives, Powering the Always-On Economy and The Asset and the Target, and building on DBA-DC-POL

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Primary audience: congressional and agency staff. Partial draft: Part Six and Findings and Asks are held for a legal reading of Section 215. The method is published; no utility, facility or site is named.

A Note on Scope

This paper is written for congressional and agency staff. It names no utility, no facility and no site, and it applies one test to every passage: whether the passage would materially help someone attack a specific facility. The method is published. The particulars that would turn a method into an operation are not.

It is the fourth paper in a family. *Generation Mix Optimization Under Competing Objectives* builds the engineering framework, *Powering the Always-On Economy* presents it to investors, and *The Asset and the Target* runs the adversarial analysis that framework left as forward work. This paper takes the same findings to the people who decide what the country requires of its infrastructure. It also builds on *DBA-DC-POL*, the policy companion in the data center series, which makes the legislative case for physical security at commercial data centers. That document is not repeated here. Where the two meet, this paper cites it and moves on.

This is a partial draft. Parts One through Five are written. The Findings and Asks section and Part Six are headed and held, for the reasons stated under each.

Findings and Asks

This section will carry the whole paper in about two pages: one paragraph on the consequence, then each ask in a sentence that names the body that must act, says whether a statute is needed, and states the trade-off in the same sentence. It is drafted last, because its asks are the recommendations of Part Four and its routing is the conclusion of Part Six, and neither is yet written.

Part One: What Must Stay On

Always-on is a matter of time

Most discussion of electricity for critical facilities starts with size: the largest loads, the biggest campuses, the fastest-growing sector. This paper starts somewhere else. A facility is always-on when losing its power harms people outside it faster than that power can be restored. The measure is time to harm, not megawatts and not sector.

That definition does real work. It puts a small hospital and a very large data center in the same conversation, and it keeps out large facilities whose outage is expensive but harms no one quickly. It also matches the direction federal policy has been moving. The Congressional Research Service describes that direction as a shift away from protecting assets toward protecting functions and the dependencies between them.

The ladder of time to harm

The corpus this paper draws on already sorts dependencies by how quickly their loss begins to hurt. The author's interdependencies mind map of January 2026 arranges restoration dependencies into five bands. The first is millisecond-critical: grid frequency regulation, protective relay response, control-system latency, 911 systems, payment clearing and time synchronization. The second covers the first four hours: activating emergency power, restoring telecommunications, keeping hospitals and emergency services running, and cooling critical facilities. The third runs from four to twenty-four hours: fuel delivery to generators, water treatment, the food cold chain and the availability of payment systems. The fourth runs from one to seven days: wastewater recovery, industrial restart and transportation repair. The fifth runs past a week: major transmission rebuilds and the replacement of large transformers.

A ladder is only useful if its rungs are measured, so the intervals in the table below come from the codes, rules and guidance that govern each function, not from recollection. Where no governing document sets a number, the text below says so.

Band	Functions placed in the band	What the governing documents say about time
Milliseconds to seconds	Frequency regulation, protective relays, control-system latency, 911 systems, payment clearing, time synchronization	Hospitals sit in the next band, but their power transfer is timed in this one: the essential electrical system must restore its life safety and critical branches within 10 seconds of losing normal power (NFPA 99 and NFPA 110, Type 10; National Electrical Code, Article 517).
Up to 4 hours	Emergency power, telecommunications restoration, hospitals and emergency services, critical cooling	A 911 communications center needs an uninterruptible supply of at least 15 minutes and 72 hours of generator fuel at full load (NFPA 1221, 2013 edition). A closed refrigerator keeps food safe for up to 4 hours (FoodSafety.gov).
4 to 24 hours	Fuel delivery to generators, water treatment, food cold chain, payment availability	Water utility backup generation typically runs 24 to 72 hours without resupply. A full, closed freezer holds about 48 hours, and a half-full one about 24 (FoodSafety.gov). Telephone central offices that serve 911 centers are certified for 24 hours of backup power, and those housing selective routers for 72 (FCC 21-99).
1 to 7 days	Wastewater recovery, industrial restart, transportation repair	The water sector design basis treats loss of potable water service beyond 72 hours as its consequence threshold, and requires restoration to minimum service for hospitals and other critical dependents within 24 hours (DBA-W).
More than 7 days	Transmission rebuild, transformer replacement	Large power transformers take twelve to twenty-four months to manufacture, and no meaningful strategic reserve stands behind them in North America

Band	Functions placed in the band	What the governing documents say about time
		(Powering the Always-On Economy, Part Four).

Two features of the table matter more than any single number in it. The first is that the intervals are short. Almost everything on the ladder is measured in hours, and the top of it in seconds, while the equipment at the bottom is measured in months. The second is how uneven the obligations are. Hospitals and 911 centers operate under explicit, enforceable time requirements. Much of the rest does not. The Federal Communications Commission's one attempt at a general backup-power minimum for the telephone network, adopted after Hurricane Katrina and set at 24 hours for central offices and 8 hours for cell sites, never took effect; the Office of Management and Budget declined to approve its reporting requirements, and the rules were removed in 2011. A function can sit high on the ladder with no rule at all telling it how long it must last.

Cold chain spans the ladder

One function does not sit on a single rung. The cold chain is the refrigerated storage and transport that keeps food, vaccines, medicines and blood products within their temperature limits from producer to patient or plate. A large frozen-food warehouse, with its thermal mass, can hold for days. A vaccine refrigerator begins to warm as soon as it loses power, and the Centers for Disease Control and Prevention treats any reading outside the storage range as an excursion that takes the product out of use until its viability is confirmed. Red blood cells must be held between 1 and 6 degrees Celsius. The mind map places the food cold chain in the four-to-twenty-four-hour band. Vaccine and blood storage belongs at the faster end.

The cold chain also fails silently. A product that warmed and cooled again can look unharmed. The harm arrives later, away from the outage that caused it, in a clinic or a kitchen that has no reason to connect the two. For a staff reader, this is the clearest case of a consequence that current outage reporting does not see.

What the ladder is for

The ladder does two jobs. It shows that data centers are one rung among many. The policy conversation of the last two years has been dominated by a single industry, and the ladder moves it back to the functions the public actually depends on, of which computing is one. It also sets up Part Three, because the rung a function sits on is the interval an accident or an adversary has to work within. A function that must be restored in seconds cannot wait for anyone to decide anything.

Part Two: Who They Serve

They serve each other first

Always-on facilities serve one another before they serve the public, and the public sits at the end of every chain. Water treatment needs power and communications. Communications needs power. The grid's own control systems need communications. Payment systems need all three. Hospitals need everything. None of this is news to anyone who works in these sectors, but it has a consequence that is easy to miss and that current regulation does not capture.

One consequence wearing many addresses

A dependency shared by many always-on facilities makes them fail together, however many of them there are. Twelve facilities behind one substation, one gas lateral or one fiber route are not twelve separate risks. They are one consequence wearing twelve addresses.

The Asset and the Target develops this point on the generation side, where it calls the quantity effective granularity. Twelve reactor modules on one control architecture have the exposure of one reactor, because an attack on the shared element defeats all twelve at once. The count that matters is not how many units there are but how many must be independently defeated. The general form of that count, taken over a declared set of shared dependencies, is carried by the design-basis method at *DBA-MA-DiD* §3.2. Staff do not need the mathematics. They need the sentence, and the observation that no current regulation counts dependencies this way. Contingency planning for the bulk power system asks what happens if the largest single element is lost. It does not ask what happens if the largest group of elements that share a dependency is lost, and for always-on load that is the question that matters.

Hard regulation and soft regulation

The sectors on the ladder are not governed alike, and the difference is best described with two terms the author set out in 2024 in work on artificial intelligence risk for the ISA Global Cybersecurity Alliance. A hard regulator is a federal or state agency that compels adherence to requirements under a penalty, usually financial and sometimes operational. A soft regulator is a federal, state or independent body that issues guidance, standards or non-enforceable rules, and that relies on the courts, typically through tort, where harm results. Standards bodies are soft regulators. They can sometimes ask a hard regulator to give their standards force.

The electric and nuclear sectors are hard-regulated. The bulk power system answers to the Federal Energy Regulatory Commission and to NERC's mandatory reliability and security standards. Nuclear plants answer to the Nuclear Regulatory Commission. Pipelines operate under security directives from the Transportation Security Administration, the current generation of which was reissued in 2026, while TSA's proposed permanent rule on surface cyber risk

management, published in November 2024, has not been finalized. The facilities now drawing the most new load from those hard-regulated systems are, for their own resilience, soft-regulated.

The private regime, and what it measures

Soft regulation is not the same as no regulation, and a staff reader who knows the industry will stop reading at any claim that data centers are unregulated. They answer to a private regime with real teeth. The Uptime Institute's Tier classification is written into tenant contracts and service-level agreements, and it bears on how lenders and insurers price a facility. Insurance brokers describe the Tier rating as a proxy for risk quality, and they report that facilities with redundant power and cooling are both easier to place and better priced. Where no public regulator reaches, the market enforces the regime through contracts and insurability.

DBA-DC-POL has already made the case that this regime does not supply a physical security standard, and that case is not repeated here. The question for this paper is different: what does the private regime measure? Tier classification examines a facility's own design. It asks whether the facility can be maintained, or can suffer a fault, without interrupting its critical load. It protects the facility's availability to its own customers, and it does that well.

It does not ask whether the facility shares a dependency with its neighbors. It does not ask what the facility's protective behavior does to the grid, which Part Three shows is now a first-order question. It does not ask what happens to the public at the far end of the chain. Its fuel expectation is modest by this paper's standards: Uptime's own guidance gives 12 hours of on-site fuel at design load as the starting point for a Tier-defined facility, the same half-day of independence the companion papers count against gas-fired generation. None of this is a criticism of the regime. It is a statement of its purpose, which is the facility and not the system.

The insurance channel is a lever as well as a gap. An underwriting requirement can move faster than a statute, and the certifying body and the insurers are parties who can act without waiting for Congress. Parts Four and Six treat them that way.

What has begun to change

The asymmetry is narrowing on one front. It concerns how large loads behave toward the grid, not how resilient they are themselves. On 4 May 2026 NERC issued a Level 3 Essential Action Alert on computational loads, the strongest advisory instrument it has, covering modeling, studies, instrumentation, commissioning, operations, protection and control, with responses due on 3 August 2026. On 9 July 2026 the Public Utility Commission of Texas approved ride-through requirements for large computational loads on the ERCOT system, effective 1 August 2026. On 16 July 2026 FERC directed NERC to file, by 31 December 2026, reliability standards on the integration of computational loads, together with registration criteria that would make some of those loads directly accountable under Section 215 of the Federal Power Act for the first time.

These are significant steps, and staff should know they are under way. They share a scope, though. Each governs the load's behavior at the point of connection: whether it rides through a disturbance, whether it is modeled, whether it is commissioned properly. None asks what the facility depends on, how long it can stand alone, or which other always-on functions fail with it. The grid-facing side is moving from soft regulation toward hard. The dependency side has not moved.

The sector said this in 2013

None of this is a new observation. When the Department of Homeland Security asked for comment in 2013 on revising the National Infrastructure Protection Plan, NERC's response placed the energy sector "at or near the top" of cross-sector cascading interdependency priorities. It warned that the revised plan had to be flexible enough for common elements to reach across sectors, and so avoid a "siloed" approach to infrastructure protection. That is the argument of this part, made by the electricity sector itself thirteen years ago.

Part Three: Why We Should Care

Whose question this used to be

Deciding what matters most across sectors, by consequence and by interdependency, was the purpose of the National Infrastructure Protection Plan and of the sector-specific plans written under it. This paper is not inventing that question. It is picking the question up where the national framework set it down, and the record of how that happened should be stated as record rather than as complaint.

The plan was last revised in 2013. The sector-specific plans written under it were last publicly archived in 2015. National Security Memorandum 22 replaced Presidential Policy Directive 21 on 30 April 2024 and remains the operative policy. In March 2025 the Department of Homeland Security terminated the Critical Infrastructure Partnership Advisory Council, effective 7 March and published on 13 March. On 18 March 2025 Executive Order 14239, *Achieving Efficiency Through State and Local Preparedness*, called for state and local governments to play a more active role in national resilience and ordered a review of critical infrastructure policy, including NSM-22. The Congressional Research Service reported in March 2026 that most critical infrastructure risk management functions were being devolved to the states, and framed the choice before Congress as reinforcing the national partnership framework or supporting devolved, state-level management.

The states do not start from nothing. Testimony before the House Energy and Commerce Committee in January 2026 recorded that, with national laboratory support, all fifty-six states and territories had completed energy security plans meeting the statutory requirements by

December 2024. Those are plans for each state's energy supply. The ranking of consequence across sectors and across state lines, which the national plan was built to hold, is a different product, and no state was given the machinery to produce it.

NERC's 2013 response makes the loss concrete. In the same filing, NERC asked that the coordinating council structure be kept intact and strengthened, and specifically that the Critical Infrastructure Partnership Advisory Council framework be preserved, because every coordinating council relied on it to protect sensitive discussions between government and industry. That is the framework terminated in March 2025. On the electricity sector's own record, the structure it asked to keep is the one that was ended. *DBA-DC-POL* treats the termination as legislative context for data center security. The point here is a wider one.

It is worth being precise about what the national framework did and did not do, because this paper does not propose to rebuild it. A 2026 analysis of the water sector plan, prepared in this corpus, puts the difference plainly. The national plan defines risk as a function of threat, vulnerability and consequence, and applies that function at the sector level to set goals and compare investments. It is probabilistic and comparative. A design basis is deterministic and set at the facility level: it states the event a facility must withstand, regardless of how likely that event is judged to be. The same analysis found that the water sector plan traced consequence only to its first level and carried no model of how quickly consequence arrives. This paper therefore adds two things the national framework never carried, a time dimension and a deterministic floor, to a question that framework once owned. The judgment about cross-sector consequence that this part makes currently has no federal owner. That absence is a consequence in its own right.

The asset and the target, in public terms

The first reason to care concerns the generation that always-on load depends on. The properties that make a generating unit most valuable to the grid are the same properties that make it the most consequential to attack. High inertia, high utilization and long fuel independence are why an optimization for low-carbon, resilient supply selects nuclear generation. They are also why an adversary's analysis selects the same unit, since its loss removes the most output and the most stabilizing inertia at once, for the longest recovery. *The Asset and the Target* shows this with a worked portfolio: the analysis that maximizes the portfolio's value maximizes, in the same act, the consequence of a single successful attack on it.

The same analysis found a cheaper second target. Four gas-fired units that look like separate, modest targets become a single target of 768 megawatts if they share one gas lateral, and they fail about twelve hours after it is cut, when the pipeline's stored gas runs out. A protection budget allocated unit by unit would spend everything on the reactor and nothing on the pipeline branch.

None of this ranking is secret. Every input is public: inertia constants, capacity and availability factors, transformer lead times, and the physics that relates them. An adversary with the same public data and an undergraduate education in power systems derives the same ranking. Declining to publish the method withholds it only from the defender, who needs it to decide where finite protection should go.

Always-on load as a source of disturbance

The second reason to care is newer, and it turns the usual framing around. Always-on loads are not only dependents of the grid. At scale, they have become a source of disturbance to it.

On 10 July 2024, in an area of the Eastern Interconnection with a high concentration of data centers, a lightning arrester failure on a 230-kilovolt line produced six successive faults in 82 seconds. About 1,500 megawatts of voltage-sensitive load transferred to backup power, and about 1,260 megawatts of it did not return for hours. Frequency rose to 60.047 hertz and voltage to 1.07 per unit before operators restored normal conditions. NERC published its incident review on 8 January 2025. The load behaved exactly as it was designed to behave, and nothing had attacked it.

It was not an isolated event. In a Level 2 Industry Recommendation issued on 9 September 2025, NERC reported that it had analyzed a series of disturbances in which more than 1,000 megawatts of large load reduced unexpectedly, most of them in 2024 or 2025, and that the rise in such events tracked the rise in large-load penetration.

Two years later the same thing happened on more than twice the scale. On 22 July 2026, a single conductor pulled free of its compression sleeve on a transmission line in the same interconnection. Data center load transferred to backup power in two waves sixty-seven seconds apart, of about 2,970 and 1,099 megawatts, and the regional transmission organization put the approximate load transferred at 3,869 megawatts, less than the sum of the two waves. Frequency peaked at 60.092 hertz. The customers returned the load to the grid within thirty minutes.

The size of that number is the point. The Eastern Interconnection's frequency response is sized each year to the two largest resource losses its balancing authorities report. For the 2026 operating year that criterion is 3,875 megawatts. One routine equipment failure removed almost exactly that much, as a common-cause response across many facilities, in the direction the criterion was never written for. The event did not break the design criterion. It revealed that nobody ever wrote one for load. The system has a scheme for losing four gigawatts of generation and no scheme at all for getting four gigawatts of load back.

That last half matters as much as the first. Backup power at these facilities is measured in minutes, so the load returns whether or not the system is ready for it. Put the same sequence inside a heat wave or a polar vortex, when the generating fleet is fully committed, and operators face a multi-gigawatt load rejection followed by an unscheduled multi-gigawatt restoration, with no dispatch authority over either and no telemetry showing how much load is sitting on batteries.

Underneath both events sits the dependency point of Part Two. Contingency analysis assumes that facilities respond independently. These facilities do not. They are built to shared conventions, with similar equipment and similar protective settings, so a single fault produces a family response. When independence fails, the largest credible event is no longer the largest single facility but the largest correlated group of them, and no one is currently computing that number. An adversary who understood the behavior would not need to touch any of the loads. It would be enough to cause the disturbance they are all set to answer.

The reliability institutions have responded, as Part Two recorded, and their response is aimed squarely at this behavior. That is the right place to start. It is not the whole problem, because a facility that rides through a fault can still share every upstream dependency with its neighbors.

Three gaps, and a fourth

The third reason to care is what the regulatory framework does not yet require. *Powering the Always-On Economy* identified three gaps and presented them to investors as risks. Stated for this audience, they are consequences.

The first is that planning reserve margins are calculated on nameplate capacity rather than on the output sources actually deliver. A region can meet its margin on paper while holding far less reserve for continuous load, which draws the same power at two in the morning as at two in the afternoon. The consequence is a shortfall that appears only when it is needed.

The second is that no rule sets a minimum fuel independence for generation serving critical load. A fleet that can run for half a day on the fuel at hand satisfies every current adequacy requirement. The consequence is that a supply interruption lasting longer than an afternoon becomes an outage for everything downstream.

The third is that no strategic reserve stands behind the large power transformers whose loss takes a facility off the system for a year or more. Physical security of critical transmission facilities is required. Recovery within a defined time is not. The Government Accountability Office found in 2023 that the Department of Energy had studied the reserve question in 2017 and again in 2022 without setting objectives, time frames or a plan to put its support into operation. The consequence is that recovery time is left to chance.

The fourth gap is the one this part opened with. No federal body now owns the judgment of which functions matter most across sectors and how their failures propagate. Every other gap assumes that someone is keeping that ranking.

Staff should hear one caution before Part Six. It is tempting to assume that the first two gaps could be closed through NERC's standards process without legislation. That is not settled, and the companion papers no longer assume it. Section 215 of the Federal Power Act excludes requirements to build new generation from the definition of a reliability standard, withholds authority over the adequacy of facilities, and preserves state authority over adequacy. A fuel-independence minimum or a reserve margin measured on delivered output may therefore fall

outside what a NERC standard can do. Part Six examines that question with the care it needs. This part does not promise a fix that Part Six may not be able to deliver.

Part Four: What Must Be Done

This part carries the recommendations. Each is stated with its limit attached, because a staff reader will quote a sentence, and a sentence quoted without its limit becomes a different recommendation. Where an ask depends on which body has the authority to act, this part says so and leaves the answer to Part Six. The legal reading on which that answer turns has not yet been confirmed.

The nuclear finding, with its qualifier

The companion papers find that nuclear generation is the only source that meets every objective their framework can express for continuous, low-carbon, resilient load. That finding stands. For this audience it appears only in one form: nuclear passes every test the framework can express, and shared dependency is a test the framework could not express until the adversarial analysis added it. A policy that encourages nuclear generation for always-on load without carrying that qualifier would reward the concentration Part Three describes. A campus of twelve reactors on one control architecture, behind one islanding boundary, has the exposure of one reactor and the price of twelve.

The limit is that the qualifier does not argue for less nuclear generation. It argues for nuclear generation held differently. Nothing here should be read as a reason to slow the licensing of new units.

Buy independence, not units

The ask is that owners of generation serving critical load declare the dependencies their units share, have that declaration reviewed by someone other than its author, compute from it how many independence classes the units actually form, meaning groups that an adversary must defeat separately, and write down a residual for every class that has no independently protected element. *The Asset and the Target* Part Seven sets out these four steps in order. The general form of the count is carried by the design-basis method at *DBA-MA-DiD* §3.2.

Two limits attach, and both matter. The first is that the declaration will be built largely from estimates. Whether two groups share a switchyard can be read from a drawing. Whether two control systems are truly independent cannot, and the honest result for a real campus is a range with the deciding judgments named, not a single number. A requirement that demands one number will get one, and it will flatter. The second is that the method is still an unpublished working draft. It now fixes the form of the report, a bounded range with the deciding judgments

named, but not yet the convention for producing one. The ask is therefore for the practice, not for a standard, until that convention is settled.

Island behind a defined boundary, and keep protection local

The ask is that always-on facilities above a threshold be able to separate from the grid behind a defined boundary and continue serving their critical functions, and that the protection at that boundary decide for itself. That means protection that acts on what it measures locally against fixed settings, on local power, without waiting for permission from a networked controller or an operator, and whose settings can be changed only by a deliberate, recorded act of a qualified person. The Asset and the Target Part Six sets out why: a protective decision that waits on outside communications or an operator will fail under the very conditions it exists for. The Predictive UFLS Ride-Through position paper adds a distinction staff should keep, at its Section 4.6. Protection that merely runs locally does not meet the ask if its decision comes from a learned model, however local the hardware. The property the ask depends on belongs to the fixed-setting protection beneath any such model. The Fuel Cell position paper adds a second tier of onsite energy measured in days, independent of the infrastructure a grid disturbance would compromise.

The limit is the one *The Asset and the Target* Part Six found by arithmetic. The islanding boundary is itself a shared dependency. A single point of connection or a single island controller returns a well-separated campus to the exposure of one unit, and a second tier fed and controlled from the same site may share the primary's fate. The ask is not for islanding capability alone. It is for boundaries that are themselves split, aligned with the independence classes, and free of a common controller. What threshold of size or time to harm should trigger the ask is not settled here.

A fuel independence standard and a ride-through duration standard

These are two related asks. The first is a minimum fuel independence for generation that serves critical load, carried from *Powering the Always-On Economy*. The second is a minimum duration of onsite stored energy at full load for critical continuous load, carried from the Fuel Cell position paper. Both are sound as engineering. A generation fleet that can run for half a day on the fuel at hand, and a critical load that can run for minutes on its batteries, are both exposed to any interruption longer than that.

The limit is authority, and it may be decisive. Section 215 of the Federal Power Act excludes requirements to build new generation from the definition of a reliability standard and withholds authority over adequacy. A fuel-duration minimum reads as an adequacy requirement and may require construction. Its justification is also of the wrong kind for Section 215: the case for it rests on the consequence to hospitals, water and other functions downstream, not on any effect the generator's fuel supply has on grid stability, so even the registration route that now reaches large loads does not reach it. It should not be assumed that NERC could adopt such a standard

without legislation until the legal reading is confirmed. Part Six routes these asks once it is. Neither should be presented to a committee as achievable by standards revision alone.

A transformer reserve with an owner and a date

The ask is a strategic reserve of large power transformers, and a recovery-time obligation beside the physical security obligation that CIP-014 already imposes on critical transmission facilities. The record shows what not to ask for. The Department of Energy reported on the reserve question in 2017 and again in 2022, and the Government Accountability Office found in 2023 that neither effort had produced objectives, time frames or an operating plan. Another study would repeat that history. The ask is for a program with a named owner, a size, a funding source and a date.

NERC's 2013 response to the national plan's request for information adds a useful detail. Recovery time objectives then lived in individual organizations' continuity plans, set and tested voluntarily by entities with mature programs. That is still broadly where recovery time sits. The limit is again authority: a recovery-time obligation framed as operation may fit within a reliability standard, while a reserve that requires procurement almost certainly needs appropriation.

Treat large loads as grid participants

The ask is that large loads carry ride-through, modeling and commissioning obligations proportionate to the disturbance they can cause. This is the one area where federal action is already under way. NERC's Level 3 alert of 4 May 2026 set expectations for how computational loads are modeled, studied, commissioned and protected. FERC's order of 16 July 2026 directed NERC to file standards and registration criteria by 31 December 2026. Texas has required ride-through for large computational loads since 1 August 2026.

The ask here is narrower than a new program. It is that Congress and FERC watch whether the standards now being written reach a facility's aggregate behavior, meaning the whole campus responding together, and the correlated behavior of many facilities built to the same conventions, rather than only the settings of individual devices. The events of July 2024 and July 2026 were family responses: no single facility misbehaved. A standard that checks each device and never the group would have passed both events. The UFLS paper's proposed change to PRC-024 is a related instrument on the generator side and should be cited as such, not folded in.

Registration is also the test of how far Section 215 reaches, and the ground on which it rests matters. Section 215 applies to users, owners and operators of the bulk power system. A load is brought within it as a user, and the obligation placed on it must be justified by the load's own effect on the fundamental reliability of the grid: what its behavior does to frequency, voltage and stability, stated to the load in those terms. That is a different ground from the United Kingdom's practice of designating an otherwise unremarkable asset as critical national infrastructure

because something critical in another sector depends on it, a practice *DBA-DC-POL* cites, correctly, as precedent for designating data centers for physical security. The two grounds should not be merged. Grid impact can support a reliability obligation under Section 215. Importance to other sectors cannot, and asks that rest on it need another authority.

The distinction exposes the hardest question in the registration route. Obligations attach to individual registered entities, but the impact in both July events was aggregate. No single facility misbehaved; many facilities built to shared conventions responded to one fault together. A single load of modest size may have no material effect on reliability, while fifty on common settings plainly do. Whether a reliability standard can place an obligation on one registered user because of what it does in combination with others, and how the statement of impact would be framed to that user, is unsettled. It is also the point at which the common-cause reasoning of *The Asset and the Target* bears most directly on federal authority. Federal jurisdiction over load interconnection is itself contested with the states, so registration should be treated as a precedent in progress rather than settled law.

The declared residual as an accountability tool

The ask is that what a facility cannot withstand be written down and made reviewable. This is the element most specific to this body of work, and the one staff are least likely to have met elsewhere. Its value is that it can be checked from outside: a facility owner who produces a dependency declaration and no residual list has not finished the analysis, and the gap is visible without access to any site. Lenders, insurers and power purchasers can require the list today without any statute, and Part Six will say so.

The limit is disclosure. A residual list is a map of what a facility cannot survive. It belongs with reviewers, lenders, insurers and regulators under appropriate protection, not in a public filing. How that protection is provided, and by whom, is part of the authority question. CIPAC was the structure that once protected sensitive discussions of this kind between government and industry, and it was terminated in March 2025.

Part Five: The Trade-offs

Every recommendation in Part Four takes something from someone. This part names who. It is written as six tensions. Each is resolved as far as the evidence allows and left open where it does not.

Who pays

Large-load interconnection costs, reserve obligations and resilience investment land either on the large load or on other ratepayers. There is no third payer. The Department of Energy's October

2025 direction to FERC proposed that large loads bear the full cost of the network upgrades their interconnection studies assign to them, and FERC's show-cause orders of 18 June 2026 put the allocation question before all six regional transmission organizations. A transformer reserve and a separated spare pool raise the same question in another form: the reserve protects every customer, but someone must fund inventory that may sit unused for years. The tension cannot be resolved by analysis. It can be made explicit, so that an allocation is chosen rather than defaulted.

Who is shed first

Under-frequency load shedding removes load in blocks, by feeder, automatically. The blocks were drawn for engineering convenience, not with always-on functions in mind, so a hospital, a water plant and a shopping district may share a feeder and a fate. Protecting always-on functions within the shedding scheme means deciding that some customers are disconnected before others. That is a political allocation, not an engineering one, and at present it is made implicitly, feeder by feeder, by whoever last revised the scheme. Part One's ladder of time to harm is one defensible basis for making it explicitly. It is not the only one, and staff should expect disagreement about it.

Onsite generation against emissions and attack surface

Onsite generation reduces an always-on facility's dependence on the grid. It also brings costs of its own. Diesel backup is the common choice, no lifecycle carbon standard applies to backup generation, and in a major outage emergency relief from environmental limits trades emissions for continuity at exactly the moment always-on load needs it. Cleaner alternatives such as batteries, fuel cells and inverter-based resources bring a different cost: each adds firmware and remote interfaces, and the companion technical paper documents how concentrated the supply chain for those devices is. A fleet of identical inverters on identical firmware is a shared dependency. Onsite resilience is worth buying. It is not free of the problem it is bought to solve.

Interconnection speed against study

The two federal actions on large loads pull in opposite directions, and staff should hear that plainly. The large-load interconnection rulemaking seeks speed: its October 2025 direction asked whether studies for loads willing to accept curtailment could be completed in 60 days. The reliability actions seek care: NERC's Level 3 alert asks for stability studies, dynamic models and commissioning processes, and FERC's July 2026 order asks for standards that will take time to write and apply. A load connected quickly and studied lightly is the load most likely to respond as a family to the next fault. Ride-through capability is also cheap when it is designed in and expensive when it is retrofitted, so every facility connected before the standards exist may join a population that must later be grandfathered or rebuilt.

Openness against security

This paper and its companions publish a method for ranking which infrastructure is most consequential to attack. Staff will be asked why. The answer is that every input to the ranking is public already: the physics, the engineering constants, the reliability statistics and the equipment lead times. An adversary with that information derives the same ranking, and declining to publish withholds it only from the defenders who must decide where finite protection goes. What is withheld is the particulars that would turn a method into an operation: named facilities, their configurations, their residual lists. The line between the two is a judgment, and this paper applies one test to every passage: whether it would materially help an attack on a specific facility. Reasonable people will draw the line in slightly different places. The case for drawing it somewhere, and publicly, is strong.

The cost of independence against concentration

In the worked example of *The Asset and the Target*, separating twelve reactor modules into three independence classes cut the worst coordinated loss to about a third, with real engineering and real money. Splitting the islanding boundary preserved that gain, and a single boundary erased it. Someone has to decide whether that purchase is required, rewarded or left to the market. Left to the market, it will be under-bought, because much of its benefit falls on the grid and on the public rather than on the owner who pays. Required, it will raise the cost of exactly the generation the country is trying to build faster. Rewarded, through insurance pricing, lender terms or recognition in interconnection and capacity rules, it may be bought where it is cheapest. The costs cannot yet be stated in dollars. They depend on site, design and vendor, and the analysis on which they rest is itself built from estimates. The direction of the trade-off is clear, and so is the principle that separation designed in costs a fraction of separation added later.

Part Six: Who Must Act

This part will sort every ask by the legal authority that can carry it, so staff can see at once which asks need Congress. It waits on a legal confirmation of the Section 215 reading summarized at the end of Part Three, and on an agreed division of labor with *DBA-DC-POL*, whose congressional instruments already cover the data center physical security case and an interconnection-conformance condition for large loads.

Sources

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Transportation Security Administration, Security Directive Pipeline-2021-01 series; notice of proposed rulemaking, "Enhancing Surface Cyber Risk Management," 7 November 2024.

T. Roxey, correspondence accompanying *A Study of AI Risks* for the ISA Global Cybersecurity Alliance, 28 October 2024, defining hard and soft regulators.

Uptime Institute, Tier Standard guidance on on-site fuel; Amwins, "Demystifying Insurance for a $7 Trillion Opportunity: Data Centers," 14 October 2025; Fullsteam Insurance, "Data Center Insurance: Coverage, Cost and Capacity in 2026," 1 September 2026.

Open Verification Items

Seven items remain open in this draft, and later versions will close them. The NERC positions quoted from the 2013 request for information are taken from the version 1 working draft of that response; the final filing in Docket DHS-2013-0024 could not be retrieved from the public docket on 15 September 2026, and the quoted language must be checked against it. The Congressional Research Service characterization of devolution to the states should be cited to the report and not to Executive Order 14239, whose text does not use the word. The 911 center requirements are from the 2013 edition of NFPA 1221, which has since been consolidated into NFPA 1225, and the current edition should be checked. No authoritative source was found for prevailing battery reserve practice at central offices and cell sites, and none is stated. Whether an obligation under Section 215 can rest on the aggregate impact of many registered users, rather than on each user's impact alone, and whether registration of loads survives the state-federal dispute over load interconnection, are both put to counsel with the Section 215 question. Part Four's statement that a fuel-independence minimum and a transformer reserve may fall outside what a NERC standard can require rests on the same unconfirmed Section 215 reading, and Part Four must be revisited when counsel answers. The 22 July 2026 figures are from the regional transmission organization's operating committee presentation and should be checked against any NERC review of the event once published.

Revision History

Version	Date	Description
0.10	17 October 2026	Brought up to date with DBA-MA-DiD version 0.2, issued on 22 September 2026. Part Four's second limit on the dependency-declaration ask no longer says that no reporting convention exists. It says the method now fixes the form of the report, a bounded range with the deciding judgments named, but not yet the convention for producing one. The source list cites version 0.2. No other change to the argument.
0.9	17 October 2026	Conformed to the corrections of 3 October 2026 in The Asset and the

Version	Date	Description
		Target and the Predictive UFLS Ride-Through position paper. The islanding ask under "Island behind a defined boundary, and keep protection local" now asks for boundary protection that decides on local measurement against fixed settings, with settings changed only by a deliberate, recorded act of a qualified person. It cites The Asset and the Target Part Six for the reason, and the UFLS paper's Section 4.6 for the distinction between protection that runs locally and protection whose decision is a learned model's. Version 0.8 was accurate on its date: the UFLS paper still requires local execution. The ask was weaker than the family's position as now stated, so this is a conformance revision and not an erratum. No other change to the argument.
0.8	17 September 2026	Publication pass for the Zenodo deposit of September 2026. The cover carries the citation block with the reserved version DOI, and the disposition line no longer describes the document as an internal draft. No change to the argument. The closing verification paragraph now says that seven items remain open in this draft rather than that they must be closed before it is distributed.
0.7	17 September 2026	Decision of 17 September 2026 on the alignment register. Part Four and Part Five use the term independence classes, as The Asset and the Target does, in place of independent groups, and Part Four defines it once in plain terms as groups an adversary must defeat separately. The paper remains a draft: Part Six and Findings and Asks are held for counsel, and the paper may be posted in that state.
0.6	17 September 2026	Alignment run across the always-on paper family. Companion papers and DBA-DC-POL are cited by title only, under the author's 17 September 2026 direction, and the family index carries versions. Parts Three and Four no longer attribute to the companion papers a claim that the fuel independence and transformer gaps can be closed without legislation, since those papers have been corrected. Part Three reports the two waves of the 22 July 2026 event beside the approximate total the regional transmission organization gives, which is less than their sum.
0.5	16 September 2026	Part Four adds the Section 215 registration point at the author's direction: a load brought within Section 215 as a user must be obligated on the ground of its own effect on grid reliability, which differs from the United Kingdom's consequence-based designation of assets that serve other sectors; the two grounds are kept separate. Part Four also records that aggregate impact across many registered users is the unsettled question in the registration route, and that the fuel-duration ask rests on downstream consequence and so falls outside it. Both questions are added to the items put to counsel.
0.4	16 September 2026	Parts Four and Five drafted from the annotated outline v0.5 and The Asset and the Target v0.10. Part Four carries seven recommendations, each with its limit, and flags every ask whose feasibility depends on the unconfirmed Section 215 reading; the independence ask carries the author's 16 September direction that dependency declarations are built from estimates

Version	Date	Description
		and should be reported as ranges. Part Five sets out the six trade-offs. The 60-day study question in the October 2025 large-load direction was verified before use. Part Six and Findings and Asks remain held.
0.3	16 September 2026	Revision history moved from after the cover to the end of the document, at the author's direction. No change to the text.
0.2	15 September 2026	Desktop archive searched after the author connected the Desktop folder. Part Three adds NERC's Level 2 Industry Recommendation of 9 September 2025, which records repeated 1,000-megawatt-plus large-load reductions in 2024 and 2025, and January 2026 House testimony on state energy security plans. The hard and soft regulator definitions remain cited to the October 2024 correspondence, because the later revision of A Study of AI Risks on the Desktop (v28) does not carry them. A draft islanding study found on the Desktop is marked not for citation without its author's permission and is not cited.
0.1	15 September 2026	First draft from the annotated outline v0.5. Parts One through Three written; Findings and Asks and Parts Four through Six headed and held. Facts rechecked on 15 September 2026 against primary sources, adding three developments the outline did not carry: the 22 July 2026 load transfer event of about 3,869 megawatts against the 3,875-megawatt Eastern Interconnection criterion, FERC's order of 16 July 2026 in Docket RD26-7-000 directing NERC to file computational-load standards and registration criteria, and the Texas approval of large computational-load ride-through requirements effective 1 August 2026. The general form of effective granularity is cited to DBA-MA-DiD §3.2, as condition three of the Project 6 transfer requires. Open verification items are listed at the end of the document.